Band-1 Hadamard: Single-Order Probe — Order 52 via Paley II over GF(25)

Human + ZoraASI full-dash (executor)

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Band-1 Hadamard: Single Order of Form $4k$ — Order 52

Honesty stamp. This note constructs and certifies **one** Hadamard matrix of order **52**. It does **not** claim the full Hadamard conjecture (Band-3 / OPEN). It does **not** construct or search order **2004** (smallest commonly cited remaining large gap after the August 2026 announcements for orders ≤ 2000).

Abstract

A prior Band-1 ledger (prime-field Paley + Sylvester + Kronecker) certified **32** Hadamard orders on box but left **52** as the unique admissible gap ≤ 64. Tonight's single lemma: implement Paley type II over the prime-power field GF(25), produce an explicit 52×52 ± 1 matrix H, and verify $\max |HH^{\mathrm{T}} - 52I| = 0$ in exact `int64` arithmetic. Normalized exhaustive search for order 52 is recorded as **intractable** (not attempted beyond a one-line abort). Full conjecture remains **OPEN**; literature smallest open large order: **2004**.

Problem (single-order Band-1)

Fixed $n = 52 = 4 \cdot 13$**:** does there exist $H \in \{\pm 1\}^{52 \times 52}$ with $HH^{\mathrm{T}} = 52I$?

- Band-1: construct / certificate-check this one order.
- Band-3: prove existence for **all** $n = 4k$ — **not tonight**.

Literature ledger (not re-proved)

Item	Status	Note
Necessary condition	$n = 1, 2$ or $4 \mid n$	classical
Paley II	order $2(q+1)$ for $q \equiv 1 \pmod 4$ prime **power**	Paley 1933
Order 52	$q = 25 = 5^2 \Rightarrow 2(25+1) = 52$	prime-power case
Prior box gap ≤ 64	**52** missing under prime-field-only toolkit	prior hadamard note
Gaps ≤ 2000 (2014 list of 12)	reported filled Aug 2026 (Alpöge et al.)	not re-verified tonight
Smallest cited open large order	**2004**	survey Sep 2026
Full conjecture	**OPEN** (Band-3)	—

Method (box)

1. **Field.** $\mathrm{GF}(25) = \mathbb{F}_5[x]/(x^2+2)$. Irreducible: x^2+2 has no root in $\mathbb{F}_5$. Identify α with $\alpha^2 \equiv 3 \pmod 5$.
2. **Quadratic character.** $\chi(0) = 0$; $\chi(a) = +1$ on the 12 nonzero squares; $\chi(a) = -1$ otherwise.

3. **Jacobsthal matrix.** $Q_{ij} = \chi(e_j - e_i)$ over an enumeration of all 25 field elements. For $q \equiv 1 \pmod 4$, $Q = Q^{\mathrm{T}}$.
4. **Conference matrix.** $C = \begin{bmatrix} 0 & \mathbf{1}^{\mathrm{T}} \\ \mathbf{1} & Q \end{bmatrix}$ of order 26.
5. **Paley II.** $H = \begin{bmatrix} C+I & C-I \\ C-I & -(C+I) \end{bmatrix}$ of order 52.
6. **Certificate.** Entries in $\{\pm 1\}$ and $\max |HH^{\mathrm{T}} - 52I| = 0$.
7. **Finite search.** Normalized backtrack for $n = 52$ aborted as intractable; $n = 4$ sanity search finds a matrix (methods check).

Results

Check	Outcome
Field irreducible	**PASS** (no roots of $x^2 + 2$ in $\mathbb{F}_5$)
#{nonzero squares}	**12**
Q symmetric	**PASS**
H shape	$(52, 52)$
± 1 entries	**PASS**
$\max \|HH^{\mathrm{T}} - 52I\|$	**0**
Wall-clock (construction+verify)	≈ 0.007 s
Exhaustive search $n = 52$	**NOT run** (intractable)
Order **2004**	**NOT constructed / NOT searched**
Full Hadamard conjecture	**OPEN**

One lemma. There exists an explicit Hadamard matrix of order 52, constructed on-box via Paley II / GF(25) and certified by $HH^{\mathrm{T}} = 52I$.

Reproducibility

- Code: `/workspace/experiments/band1/hadamard_order52/hadamard_52_paley_II_gf25.py`
- Results: `results.json`, `SUMMARY.txt`, `H52_pm1_bits.json`
- Companion ledger: `papers/2026-09-11_Band1_Computational_Experiment_hadamard.md`

Conclusion

Band-1 deliverable: order **52** constructed and verified. Closes the prior on-box ≤ 64 methods gap. Does **not** settle the Hadamard conjecture; does **not** touch order **2004**.